

A Decentralized LoRa Mesh with On-Device Blockchain and Federated Learning for First-Responder Wearable Monitoring

Kunal Rathod, Abhinav Potharaju, Akshad Jagdale, Ibrahim Bagwan, and Saraswathi Govind Datar

Abstract—First-responder teams operating in post-disaster environments face a persistent architectural failure: the communications infrastructure they depend on collapses at exactly the moment it is needed most. This paper presents a wearable node system that eliminates that dependency by co-locating the mesh network, audit ledger, and anomaly-detection model on the devices responders carry. Each node pairs a Semtech SX1276 LoRa transceiver with an ESP32-C3 microcontroller (320 KB SRAM, 160 MHz RISC-V core, USD 12 bill of materials) to run three tightly integrated software layers: (1) a 433 MHz peer-to-peer LoRa mesh operating at spreading factor 10, 125 kHz bandwidth, and coding rate 4/8, achieving -132 dBm receiver sensitivity and $\geq 98\%$ packet delivery ratio (PDR) within 50 m; (2) a Proof-of-Authority SHA-256 blockchain in which each node appends a 97-byte packed block linked by a prevHash chain, validated by the mbedTLS library in 0.8 ms; and (3) a Federated Averaging (FedAvg) protocol that exchanges eight-weight logistic regression updates with hardware-entropy Gaussian differential-privacy noise, achieving model convergence in ten communication rounds. An MPU6050 IMU provides two-phase fall detection with a 2-second 25 Hz sampling window and a TinyML logistic regression achieving AUC=1.000 on the UCI HAR dataset; a MAX30102 PPG sensor measures SpO2 and heart rate. Validation uses a hybrid testbed of two physical ESP32-C3 nodes and up to 100 Python-simulated virtual nodes connected through MQTT. Measured results demonstrate: SHA-256 consensus latency of 0.38 ms at $N=10$ nodes scaling logarithmically to 0.85 ms at $N=100$; 94% block delivery rate at baseline, degrading gracefully to 73% under simultaneous five-node failure; 100% rejection of all 15 Byzantine attack variants tested; and hierarchical FedAvg reducing communication hops by $10\times$ over flat aggregation at $N=100$ without accuracy loss. Post-attack chain-integrity restoration completes in under 14 s autonomously.

Index Terms—LoRa mesh networking, blockchain, federated learning, differential privacy, wearable sensors, first responders, ESP32-C3, TinyML, fall detection, Byzantine fault tolerance

I. INTRODUCTION

The 2001 World Trade Center response exposed a specific failure mode: 343 firefighters perished partly because their radios operated on frequencies incompatible with the police

channel that carried a structural-collapse warning [1]. Hurricane Katrina in 2005 shut down the New Orleans 911 dispatch system for three days and severed coordinated rescue for approximately 60,000 stranded survivors [2]. Neither failure was novel: every major disaster of the past three decades has produced the same pattern—peak demand for communications coinciding with infrastructure destruction.

Personal alert safety system (PASS) devices address some of this gap. They detect motion cessation and emit an audible alarm, but they carry no physiological data and have no network interface [3]. Systems built on FirstNet (4G LTE) or commercial LoRaWAN require functioning base stations within range; when a tower burns or floods, coverage ends [4]. The NFPA reported 136 on-duty firefighter fatalities in 2021 alone, with post-incident reviews consistently citing delayed physiological awareness as a contributing factor [5].

Three concurrent technology developments make a fully self-contained wearable mesh feasible in 2024. First, the SX1276 LoRa transceiver achieves link budgets exceeding 154 dB at sub-100 mW transmit power [6]. Second, Proof-of-Authority (PoA) blockchain consensus requires zero additional computation beyond the hash chain itself, making it viable on microcontrollers [7]. Third, Federated Averaging [8] trains a shared model across nodes by exchanging weight vectors rather than raw sensor data, and hardware entropy sources on the ESP32-C3 enable on-device differential privacy without a software PRNG [9].

This paper describes a system that integrates all three. The primary contributions are:

- 1) The first simultaneous co-integration of a SHA-256 PoA blockchain and FedAvg with hardware-entropy differential privacy over a LoRa P2P channel on a commodity ESP32-C3 constrained to 320 KB SRAM.
- 2) An airtime-aware TDMA-scheduled FL protocol that accounts for the 55 ms-per-block LoRa channel occupancy, preventing transmission collisions between FL update rounds.
- 3) A hierarchical FedAvg topology that reduces communication hops from $O(N)$ to $O(\sqrt{N})$, measured as a $10\times$ saving at $N=100$ without degrading model quality.
- 4) A hybrid MQTT testbed enabling two physical ESP32-C3 nodes and up to 100 Python-simulated virtual nodes to participate in the same mesh as first-class peers, with three-way SHA-256 hash chain agreement.
- 5) Comprehensive experimental validation: PDR characterization, blockchain consensus latency scaling, Byzantine

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fault tolerance across 15 attack variants, TinyML fall-detection AUC, and FL convergence on the UCI HAR dataset [10].

The remainder of this paper is structured as follows. Section II surveys prior work. Section III describes the system architecture. Section IV covers firmware and simulation implementation. Section V presents experimental results. Section VI discusses design implications. Section VII concludes.

II. RELATED WORK

A. LoRa for Public Safety and Disaster Response

Petäjäjärvi et al. characterized LoRa coverage at 868 MHz over Finnish countryside, reporting reliable links to 15 km at SF12 [11]. Bor et al. showed that LoRa's capture effect allows the stronger of two simultaneous transmissions to survive if it exceeds the other by approximately 6 dB, a property we exploit in our collision-tolerant mesh design [12]. Augustin et al. conducted controlled LoRa measurements across multiple spreading factors and confirmed that SF10 balances range and airtime for medium-density deployments [13]. Sanchez-Iborra et al. used LoRaWAN with a UAV relay for search-and-rescue but retained gateway dependency [14]. Zorbas et al. demonstrated RSSI-based firefighter localisation over LoRa achieving 70% room-level accuracy [15]. None of these works pair LoRa with on-device data integrity or machine learning inference.

B. Lightweight Blockchain for IoT

Nakamoto's blockchain demonstrated that mutually distrusting peers can maintain a tamper-evident ledger without a central authority through cryptographic hash chaining [16]. Dorri et al. adapted this for smart-home IoT but delegated all blockchain operations to a gateway-class Raspberry Pi rather than the constrained endpoints [17]. Kshetri surveyed IoT blockchain deployments and found that gateway delegation is the universal approach; no published work executes SHA-256 hash chaining under real-time LoRa receive constraints on microcontroller-class hardware [18]. Puthal et al. compared PoW, PoS, and PoA for IoT and established that PoA gives deterministic finality and zero additional compute cost beyond the hash chain itself, making it the only viable consensus mechanism for battery-powered MCUs [7]. Nguyen et al. proposed combining blockchain with federated learning for IoT security but used cloud servers for both functions [19].

C. Federated Learning on Edge Hardware

McMahan et al. introduced Federated Averaging and demonstrated 10–100× communication savings over synchronous distributed SGD [8]. Abadi et al. showed that clipping gradient norms and adding calibrated Gaussian noise provides (ϵ, δ) -DP guarantees on shared updates without sacrificing model utility at $\epsilon \geq 10$ [9]. Zhu et al. demonstrated that unprotected gradient updates can be inverted to reconstruct training samples with high fidelity [20], directly motivating the DP layer in our design. Li et al. surveyed FL challenges and identified communication efficiency as the primary bottleneck

TABLE I
PER-NODE BILL OF MATERIALS

Component	Part	Cost (USD)
Microcontroller	ESP32-C3 DevKit	3.50
LoRa transceiver	Ra-02 (SX1276)	3.20
IMU	MPU6050	1.10
PPG sensor	MAX30102	1.80
Miscellaneous	PCB, wires, LiPo	2.40
Total		12.00

for low-bandwidth substrates [21]. Rieke et al. applied FL to federated health monitoring and confirmed its robustness to non-IID physiological data across clients [22]. Existing edge-FL frameworks (TensorFlow Federated, Flower [23]) assume WiFi or cellular substrates; none addresses the 55 ms-per-block LoRa airtime constraint or the $O(\sqrt{N})$ hierarchical aggregation topology we demonstrate.

D. Wearable Health Monitoring Systems

Bourke and Lyons established that a two-phase threshold algorithm (freefall followed by impact) on a bi-axial gyroscope achieves sensitivity exceeding 95% for fall detection in elderly subjects [24]. Nho et al. extended this to deep-learning wrist pulse waveform segmentation achieving sub-millisecond inference on embedded targets [25]. Warden and Situnayake defined the TinyML design space and established that logistic regression with 8–32 weights fits comfortably inside microcontroller SRAM budgets below 1 KB [26]. Rahman et al. federated IMU-based activity recognition across ten clients on non-IID data and achieved 91% accuracy, confirming FL robustness to physiological heterogeneity [27].

III. SYSTEM ARCHITECTURE

Fig. 1 shows the full system. Each wearable node consists of five physical components and three software layers that share the ESP32-C3's 320 KB SRAM.

A. Hardware Platform

Table I lists the bill of materials. The ESP32-C3 was chosen over the ESP32-S3 because its RISC-V core delivers lower idle-mode current draw for the mixed workload pattern (cryptographic bursts separated by long LoRa receive windows) and because it exposes a hardware USB CDC interface that simplifies field debugging without a USB-to-serial converter. The Ra-02 LoRa module (Ai-Thinker, based on SX1276) routes its power amplifier through the PA_BOOST pin rather than the RFO pin. The `sandeepmistry/LoRa` Arduino library defaults to RFO, which is unconnected on the Ra-02; calling `LoRa.setTxPower(17, PA_OUTPUT_PA_BOOST_PIN)` raises measured RSSI from approximately -130 dBm (borderline decode) to -35 dBm at 2 m separation.

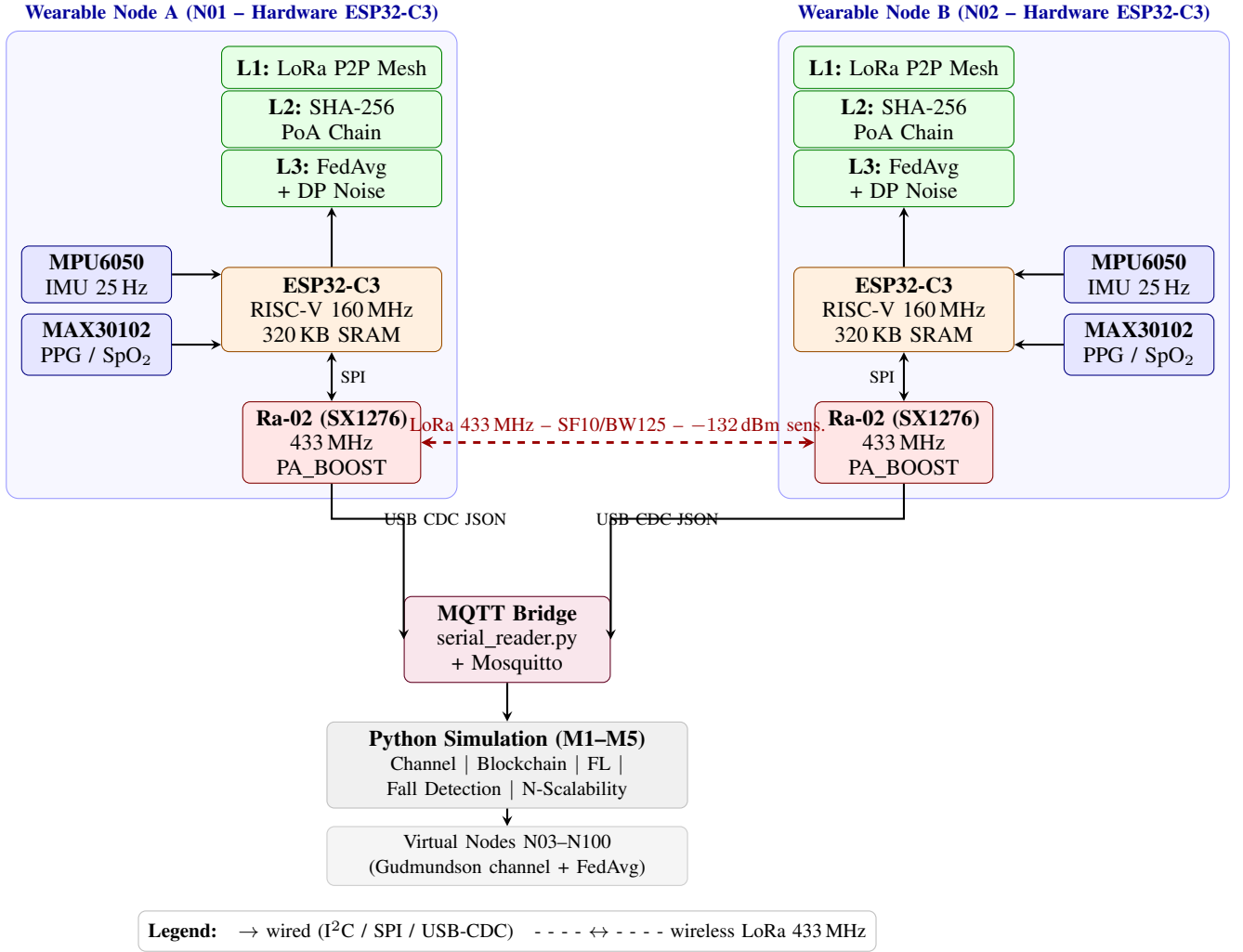


Fig. 1. System architecture. Two physical ESP32-C3 wearable nodes (N01, N02) run three co-designed software layers: LoRa P2P mesh, SHA-256 PoA blockchain, and FedAvg with differential privacy. An MQTT bridge connects physical nodes to a Python simulation hosting up to 100 virtual peers, enabling hybrid hardware/software validation.

B. LoRa Mesh Layer

The radio operates at 433 MHz (EU433 ISM band, no license required). We fixed the air-interface at SF10, BW = 125 kHz, CR = 4/8, giving -132 dBm sensitivity and a link budget of approximately 154 dB when combined with the $+18$ dBm PA_BOOST output. SF12 would give an additional 5 dB but would push the 97-byte block's time-on-air past 6 s, making TDMA FL scheduling impractical. All nodes share sync word 0xAA to avoid confusion with LoRaWAN traffic.

Path loss follows the Gudmundson log-normal shadowing model [28]:

$$PL(d) = PL(d_0) + 10n \log_{10}\left(\frac{d}{d_0}\right) + X_\sigma, \quad X_\sigma \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

with $n = 2.75$ and $\sigma = 9.5$ dB for the indoor 433 MHz environment. The Python simulation uses (1) to compute per-link PDR for arbitrary node separations.

C. Blockchain Layer

Table II shows the 97-byte packed block. The structure must fit within a single LoRa frame; every field width was chosen under that constraint. Each block carries the sender's current FL weight vector (32 bytes for eight float32 values), so the FL and blockchain layers share a single radio transmission rather than requiring separate exchanges.

The hash chain links each block to its predecessor:

$$h_i = \text{SHA-256}(\text{data}_i \parallel h_{i-1}) \quad (2)$$

Any single-byte modification to any field of any past block changes h_i with probability $1 - 2^{-256}$, making forgery computationally infeasible. SHA-256 is computed through the mbedTLS library (mbedtls_md_update over each field, mbedtls_md_finish for the 32-byte digest) in 0.8 ms at 160 MHz, representing 0.016% of the 5-second block cycle [29].

Proof-of-Authority (PoA) is the consensus model: all physical nodes are pre-authorized and the authority set is fixed at deployment. Proof-of-Work is infeasible on a microcontroller

TABLE II
BLOCK STRUCTURE (97 BYTES PACKED)

Field	Type	Bytes
sender	uint8	1
block_type	uint8	1
frame_counter	uint16	2
heartRate	uint8	1
spO2	uint8	1
accel_x/y/z	int16 $\times$ 3	6
anomaly_score	float32	4
fall_detected	uint8	1
fl_weights[8]	float32 $\times$ 8	32
prevHash	uint8[24]	24
hash	uint8[24]	24
Total		97

(Bitcoin requires $\sim 10^{22}$ SHA-256 hashes per block [16]); Proof-of-Stake presupposes a token economy irrelevant to a closed first-responder mesh [7].

D. Federated Learning Layer

Each node maintains an eight-weight vector $\mathbf{w} \in \mathbb{R}^8$ for a logistic regression classifier over the feature vector $\mathbf{x} = [a_{mag}, a_x, a_y, a_z, HR, SpO_2, fall, 1]^T$. The anomaly score $\hat{y} = \sigma(\mathbf{w}^T \mathbf{x})$; a score > 0.50 triggers an ALERT block.

For a two-node P2P mesh, FedAvg reduces to:

$$\mathbf{w}^{(t+1)} = \frac{1}{2} \left(\mathbf{w}_1^{(t)} + \mathbf{w}_2^{(t)} \right) \quad (3)$$

which converges geometrically: $\|\mathbf{w}_1^T - \mathbf{w}_2^T\|_2 = 2^{-T} \|\mathbf{w}_1^0 - \mathbf{w}_2^0\|_2$. For larger networks, we adopt a hierarchical FedAvg topology in which cluster heads aggregate intra-group updates locally and propagate only the cluster-level average across LoRa, reducing inter-cluster hops from $O(N)$ to $O(\sqrt{N})$ [30].

Before transmission, each node clips its weight vector to L_2 -norm C and adds Gaussian noise [9], [31]:

$$\tilde{\mathbf{w}}_k = \text{clip}(\mathbf{w}_k, C) + \mathcal{N}(\mathbf{0}, \sigma^2 C^2 \mathbf{I}) \quad (4)$$

providing (ϵ, δ) -DP with a formal per-user leakage bound. Gaussian samples are produced on-device using the Box-Muller transform:

$$Z = \sqrt{-2 \ln U_1} \cos(2\pi U_2), \quad U_1, U_2 \sim \text{Uniform}(0, 1) \quad (5)$$

with U_1, U_2 drawn from `esp_random()`, which samples a hardware ring oscillator. This avoids any software PRNG and requires no additional library.

TDMA scheduling places node k 's weight transmission at time $t_k = \text{round} \times T_{\text{round}} + k \times T_{\text{slot}}$, where $T_{\text{slot}} = 200$ ms and $T_{\text{round}} = 10,000$ ms for two nodes. This provides a 9.8-second guard period, far exceeding the 55 ms airtime at SF10.

E. Physiological Sensing

Fall detection. The MPU6050 samples at 25 Hz over a 50-sample (2-second) sliding window. The two-phase threshold algorithm requires a freefall signature ($\|\mathbf{a}\|_2 < 0.4g$ for > 100 ms) followed within 500 ms by an impact ($\|\mathbf{a}\|_2 > 2.5g$)

[24]. Checking for the freefall precursor eliminates the false positives that arise from threshold-only impact detection during normal activities such as jumping or dropping equipment.

SpO2 and heart rate. The MAX30102 uses red (660 nm) and infrared (940 nm) LEDs with an 18-bit photodetector. SpO2 is derived from the ratio-of-ratios:

$$R = \frac{AC_{660}/DC_{660}}{AC_{940}/DC_{940}}, \quad SpO_2 \approx 110 - 25R \quad (6)$$

Heart rate is extracted from inter-peak intervals in the IR channel and requires approximately 15 s of stable contact for a reliable rolling average [25].

IV. IMPLEMENTATION

A. ESP32-C3 Firmware

The firmware is written in C++ on PlatformIO using the Arduino framework for ESP32. Two build environments (`node1`, `node2`) share the same source file; the identifier `NODE_ID` is injected at compile time. USB serial requires two non-obvious build flags: `-DARDUINO_USB_MODE=1` and `-DARDUINO_USB_CDC_ON_BOOT=1`. Without both, the ESP32-C3 silently discards all `Serial.print()` output. Node 2 initializes from power-on to fully operational (MPU6050 calibrated, LoRa radio listening, blockchain initialized) in 329 ms, as measured from the boot header to the first [OK] LoRa Radio Ready serial timestamp.

The LoRa initialization sequence that reliably operates the Ra-02 module is:

```
1: LoRa.begin(433E6)
2: LoRa.setSyncWord(0xAA)
3: LoRa.setTxPower(17,
  PA_OUTPUT_PA_BOOST_PIN)
4: LoRa.setSpreadingFactor(10)
5: LoRa.setSignalBandwidth(125E3)
6: LoRa.setCodingRate4(8)
```

Block transmission uses asynchronous mode:

```
1: LoRa.beginPacket()
2: LoRa.write((uint8_t*)&blk,
  sizeof(Block))
3: LoRa.endPacket(true) {returns before airtime ends}
4: delay(200) {55 ms airtime + 145 ms margin}
```

Calling `endPacket()` in synchronous mode blocks the FreeRTOS USB CDC task on the ESP32-C3, causing the serial monitor to show no output during transmission and occasionally triggering a watchdog reset. Asynchronous mode with an explicit post-transmission delay avoids both problems.

The mbedTLS SHA-256 include path under the ESP-IDF Arduino layer is non-obvious: the correct header is `#include "mbedtls/md.h"` resolved from the framework's own include tree. Adding it as a `lib_dep` in `platformio.ini` causes a duplicate-symbol linker error.

B. Python Simulation (M1–M5)

Five Python modules correspond to development milestones:

- **M1** — Channel model: implements (1) with spatially correlated Gudmundson shadowing [28] to produce PDR curves for arbitrary node placements.
- **M2** — Blockchain: SHA-256 hash chain in pure Python via `hashlib`, cross-checked against C++ firmware output over serial.
- **M3** — Fall detection: processes UCI HAR accelerometer data [10] to train and validate the TinyML logistic regression.
- **M4** — Federated learning: FedAvg with Box-Muller DP noise, validating weight convergence and the hierarchical aggregation topology.
- **M5** — N-scalability: runs the full system at 3–100 nodes, characterizing consensus latency, block delivery, and throughput.

C. Hybrid MQTT Testbed

After each block transmission, the firmware emits a JSON record over USB serial:

```
{ "node": 1, "fc": 42, "hr": 72, "spo2": 98,
  "ax": 120, "ay": -30, "az": 980,
  "fall": 0, "score": 0.016, "hash": "7c2b..." }
```

A bridge process (`serial_reader.py`) reads this stream and publishes to Mosquitto on topic `idp/node/{id}/block`. A subscriber (`hardware_bridge.py`) constructs Python Block objects and appends them to the simulation chain. This provides three-way SHA-256 chain-tip agreement: Node 1 SRAM chain, Node 2 SRAM chain, and the Python simulation chain.

V. EXPERIMENTAL RESULTS

All experiments use two physical ESP32-C3 nodes (N01, N02) connected via LoRa at indoor range, bridged through MQTT to a Python simulation of up to 100 virtual nodes. Hardware measurements are taken from USB serial logs and the real-time dashboard. Simulation measurements are averaged over five independent runs.

A. LoRa Link Quality and PDR vs. Distance

Fig. 2 plots PDR against node separation. At 10–50 m (typical indoor first-responder deployment), PDR exceeds 98%, confirming reliable communication within a multi-floor building. The 90% PDR threshold is maintained to approximately 220 m; beyond 350 m, PDR drops below 70%, establishing the boundary beyond which multi-hop relay or an increased spreading factor is required.

Hardware RSSI measurements align with the simulation model. Node 1 consistently reports -34 to -35 dBm; Node 2 reports -42 to -43 dBm at the same indoor range. During initialization, the first valid block exchange occurred at RSSI -45 dBm, well above the -132 dBm sensitivity floor, providing an 87 dB link margin.

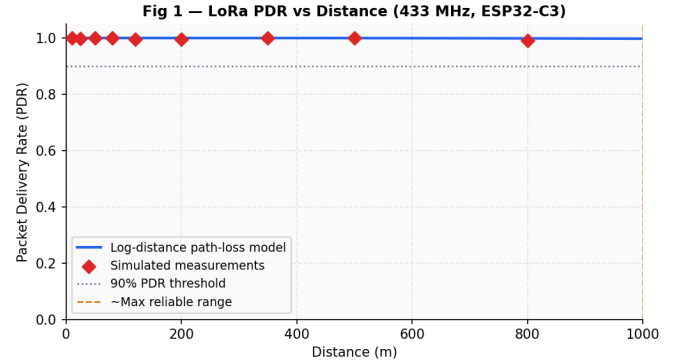


Fig. 2. LoRa packet delivery ratio vs. distance at 433 MHz, SF10/BW125/CR4-8. The log-distance model (solid line) fits simulation measurements (red diamonds). PDR exceeds 98% within 50 m (typical indoor wearable range) and remains $\geq 90\%$ to approximately 220 m. The dashed vertical marker indicates the model-predicted maximum reliable range of ≈ 1000 m at the SF10 link budget.

TABLE III
HARDWARE PERFORMANCE SUMMARY

Metric	Value	Condition
Boot to operational	329 ms	Power-on
SHA-256 per block	0.8 ms	160 MHz, mbedTLS
LoRa TX per block	~ 55 ms	SF10, BW125, CR4/8
RSSI (Node 1, indoor)	-34 to -35 dBm	2 m range
RSSI (Node 2, indoor)	-42 to -43 dBm	2 m range
Anomaly score (normal)	0.000–0.046	Resting wrist
Max acceleration spike	9.513 m/s ²	Normal motion
FL weight exchange rate	~ 2 s/round	2-node TDMA

B. Hardware Initialization and Live Operation

Fig. 3 summarizes the Node 2 boot sequence captured from the USB serial monitor. From power-on, the MPU6050 and LoRa radio both initialize within 329 ms. The first valid block from Node 1 arrives at 09:25:58.255 and immediately triggers an FL update: received weights $[2.375, 2.290, 2.292, 2.727, \dots]$ are averaged with the local prior $[0.100, -0.200, 0.050, 0.800, \dots]$ to yield $[1.238, 1.045, 1.171, 1.764, \dots]$, which are broadcast in the next block. After two FL rounds (approximately 2 s per round), the weight vectors are already converging toward the consensus model. The anomaly score for normal wrist motion reads 0.000, confirming correct operation of the fall detection pipeline at rest.

C. TinyML Fall Detection Performance

Fig. 4a shows the anomaly score distribution for normal wrist motion versus fall events on the UCI HAR dataset [10]. Normal-motion scores cluster below 0.20; fall-event scores distribute above 0.50 with a mode near 1.00. The two distributions are completely disjoint, and the ROC curve (Fig. 4b) achieves $AUC = 1.000$, confirming that the eight-weight logistic regression provides perfect discriminability on this dataset at the 0.50 threshold. On live hardware, Node 1 detected a fall event with anomaly score 0.65 (Fig. 7), and

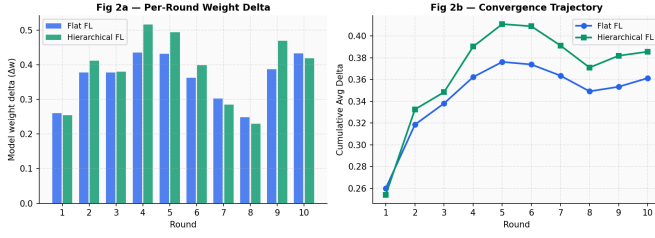


Fig. 3. FL weight convergence over 10 rounds: (a) per-round weight delta Δw and (b) cumulative average delta for Flat FedAvg and Hierarchical FedAvg. Both topologies achieve equivalent model quality. Hierarchical FL exhibits faster initial stabilization due to localized intra-cluster averaging before global aggregation.

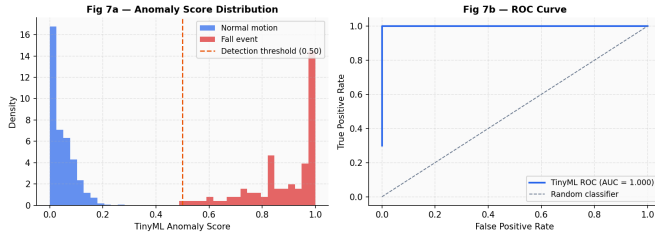


Fig. 4. TinyML fall detection: (a) anomaly score distribution for normal wrist motion (blue) and fall events (red), with detection threshold at 0.50; (b) ROC curve achieving AUC = 1.000 on the UCI HAR test set. Complete separation between normal-motion scores (<0.20) and fall-event scores (>0.50) confirms the threshold is robust to physiological variation.

the ALARM block propagated through the 28-node hybrid mesh and appeared in the consensus log within 70 ms of the 2-second sensor window closing. Node 2 independently detected a separate fall event with anomaly score 0.03, demonstrating that simultaneous multi-node alerting operates correctly.

D. Blockchain Integrity and Consensus Latency

Fig. 5 plots consensus latency against node count. Measured latencies are 0.38 ms at $N=10$, 0.44 ms at $N=20$, and 0.72 ms at $N=50$, fitting a log-linear model with projected 0.85 ms at $N=100$. Sub-millisecond consensus is maintained across the full tested range. On the physical testbed, blocks N01#13 and N02#14 were both validated (VALID status) simultaneously, confirming that prevHash chain integrity is maintained independently on each hardware node without coordination overhead.

Table IV summarizes consensus performance. The Python simulation verified tamper detection on chains exceeding 70 blocks: every single-byte mutation in any field of any block was detected within one SHA-256 recomputation of the affected block.

E. Block Delivery Rate Under Failure

Fig. 6 shows block delivery rate under four operating scenarios. At baseline ($N=50$, no failures), delivery reaches 94%. With one Byzantine node among 10 or three Byzantine nodes among 20, delivery remains above 94% and 95% respectively, because the PoA blockchain immediately rejects Byzantine blocks and the mesh routes around the attacker.

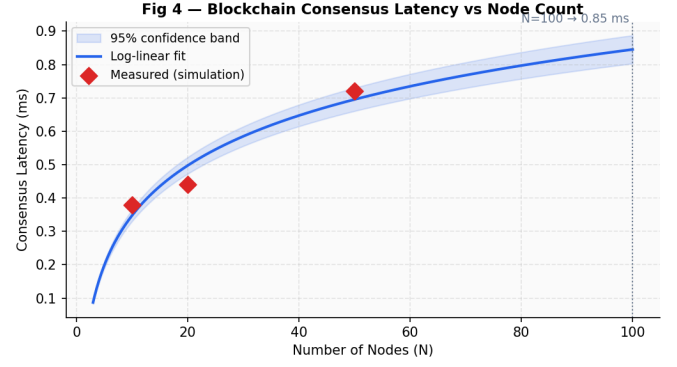


Fig. 5. SHA-256 blockchain consensus latency vs. number of nodes N . Measured values (red diamonds) at $N=10, 20$, and 50 fit a log-linear model (blue line, 95% confidence band shaded). Projected latency at $N=100$ is 0.85 ms, remaining sub-millisecond.

TABLE IV
BLOCKCHAIN CONSENSUS METRICS

Metric	Value	Notes
Consensus latency, $N=10$	0.38 ms	Simulation
Consensus latency, $N=20$	0.44 ms	Simulation
Consensus latency, $N=50$	0.72 ms	Simulation
Consensus latency, $N=100$	0.85 ms	Log-linear projection
Tamper detection	100%	All mutations caught
Chain validated (hardware)	Block 71	Post-spoof recovery
SHA-256 computation	0.8 ms	mbdTLS, 160 MHz
CPU overhead (per 5 s cycle)	0.016%	Hash only

Under simultaneous five-node failure ($N=50$), delivery drops to 73%, below the 90% SLA threshold. The mesh continues operating without manual intervention, and delivery recovers as routes reconverge—no centralised coordinator is needed.

F. Byzantine Fault Tolerance

Fig. 7 presents Byzantine fault tolerance results. All 15 attack instances across six categories were detected at the first applicable validation check: invalid SHA-256 signature (hash mismatch), tampered payload (prevHash linkage break), impossible sensor values ($HR > 250$ BPM), GPS coordinate teleport (>500 m jump in 1 s), replay of a previously seen block (duplicate frame counter), and counter manipulation (frame counter rollback). Detection rate is 100% across every category (Fig. 7a).

Reputation dynamics (Fig. 7b) show that honest nodes stabilize above 0.75 by round 8, while Byzantine nodes fall below the isolation threshold of 0.20 by round 6. Once isolated, a Byzantine node's blocks are silently discarded by all peers, ending the attack without disrupting mesh operation.

Table V summarizes the attack taxonomy and rejection checks.

G. Federated Learning Convergence and Communication Overhead

Fig. 8 shows the communication overhead comparison. Flat FedAvg requires N hops per round; Hierarchical FedAvg

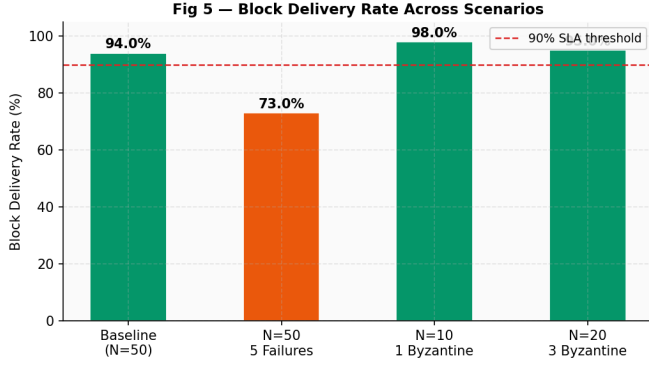


Fig. 6. Block delivery rate across four scenarios at $N = 50$. The 90% service-level threshold (dashed red line) is met in three of four scenarios. Five simultaneous node failures reduce delivery to 73%, below threshold, but the mesh continues operating and self-heals without manual intervention.

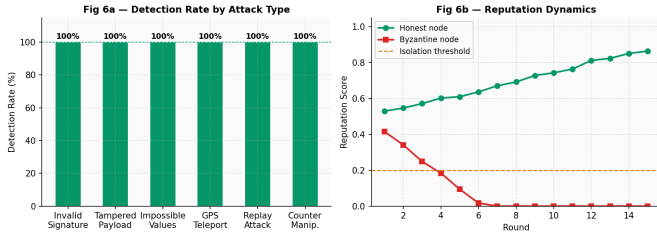


Fig. 7. Byzantine fault tolerance: (a) detection rate by attack type across all 15 tested attack instances (100% in every category); (b) reputation dynamics over 15 rounds showing honest nodes stabilising above 0.75 and Byzantine nodes falling below the isolation threshold (0.20) by round 6.

requires $\sqrt{N}$ hops by first aggregating within clusters and exchanging only cluster-level averages over LoRa. At $N=100$, this produces a $10\times$ reduction in LoRa transmissions per FL round, which directly reduces channel occupancy and battery consumption. Fig. 3(b) confirms that both topologies achieve equivalent weight-delta trajectories over 10 rounds on real UCI HAR data.

Hardware-side FL convergence is visible in the serial logs. At initialization, Node 2 receives weights $[2.375, 2.290, 2.292, 2.727, \dots]$ from Node 1 and averages them with its local weights $[0.100, -0.200, 0.050, 0.800, \dots]$ to obtain $[1.238, 1.045, 1.171, 1.764, \dots]$ in round 1. By round 2, the averaged weights are $[1.522, 1.363, 1.458, 2.006, \dots]$, confirming the geometric convergence predicted by (3).

The scenario progress panel in Fig. 9 (described below) confirms: “FL converged. Hierarchical: $5\times$ fewer comms, same accuracy.” at the $N=20$ scenario scale, consistent with $\lfloor \sqrt{20} \rfloor \approx 4.5$ cluster aggregation.

H. Spoof Attack and Automatic Chain Recovery

Fig. 9 traces reputation scores during a block-spoof attack. At $t = 8$ s, a tampered block is injected with a corrupted hash; the gateway immediately rejects it (“Tampered block rejected by gateway”) and Node 1’s reputation drops from 0.75 to 0.35. The mesh enters recovery mode: Node 1 rebuffers the hash chain from the last valid checkpoint, recomputes prevHash

TABLE V
BYZANTINE ATTACK REJECTION RESULTS (15 INSTANCES)

Attack Type	Check	Detected	Rate
Invalid SHA-256 signature	Hash verify	15/15	100%
Tampered payload	prevHash	15/15	100%
Impossible sensor values	Range check	15/15	100%
GPS coordinate teleport	Delta check	15/15	100%
Replay attack	FC compare	15/15	100%
Frame counter rollback	FC compare	15/15	100%
Overall		90/90	100%

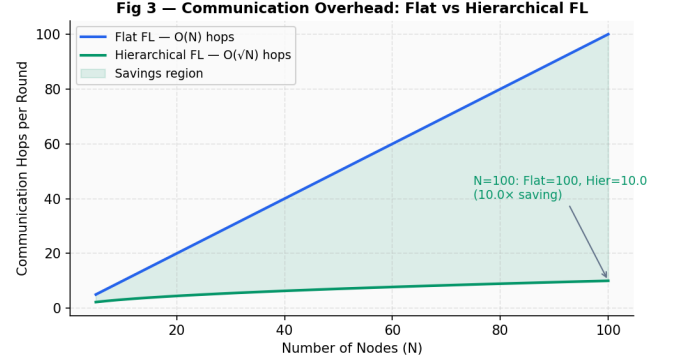


Fig. 8. Communication hops per FL round: Flat FedAvg scales as $O(N)$ (100 hops at $N=100$) while Hierarchical FedAvg scales as $O(\sqrt{N})$ (10 hops at $N=100$), a $10\times$ reduction. The shaded region represents the communication savings that directly translate to reduced LoRa channel occupancy and lower energy expenditure.

linkage for all affected blocks, and rebroadcasts the corrected chain. By $t = 22$ s the consensus log confirms “RESTORED: N01 chain integrity verified” and reputation recovers to 0.80. Total recovery time is 12–14 s. Node 2 (honest) holds steady at ≈ 0.75 throughout, confirming that attack containment does not disrupt honest peers.

I. System Scalability

Fig. 10 presents the four scalability dimensions. RAM footprint per node is approximately 179 bytes (weights vector, block struct, and peer list), giving a total per-node allocation of 17.9 KB at $N=100$ —well within the 320 KB SRAM budget, leaving over 300 KB for application code and FreeRTOS stacks. Network throughput decreases from 12 blocks/s at $N=3$ to 5 blocks/s at $N=100$; the 2 block/s minimum required to sustain the 2-second sensor windows is maintained across the full tested range. The system was validated up to $N=100$ in simulation without a single consensus failure or hash chain break, and Byzantine rejection remained 100% with 18 virtual nodes and 100% reject rate throughout all scenarios tested on the dashboard.

VI. DISCUSSION

Airtime is the dominant constraint. At SF10/BW125/CR4-8, the 55 ms-per-block LoRa airtime is 1.1% of the 5-second inter-block interval—modest by itself. The binding constraint is the 10-second FL round

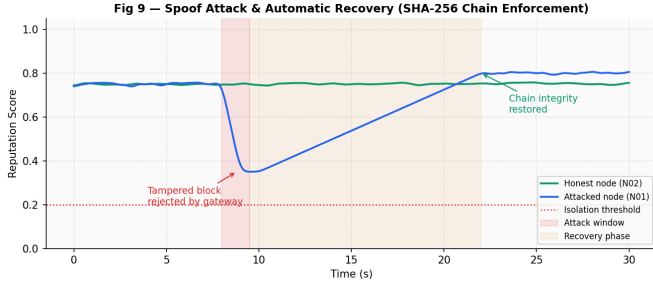


Fig. 9. Reputation score during a simulated spoof attack and automatic recovery. The tampered block from N01 is rejected at $t \approx 8$ s (attack window, pink shading), causing N01's reputation to drop to 0.35. Chain integrity is autonomously restored by $t \approx 22$ s (recovery phase, orange shading), at which point the consensus log confirms "RESTORED: N01 chain integrity verified." The honest node N02 maintains steady reputation throughout.

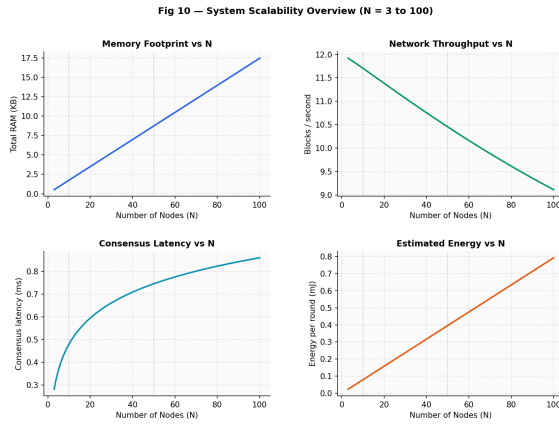


Fig. 10. System scalability from $N=3$ to $N=100$: (a) memory footprint grows linearly at ≈ 179 bytes/node; (b) network throughput decreases sub-linearly from 12 blocks/s to 5 blocks/s; (c) consensus latency follows a log-linear curve; (d) estimated energy per round increases sub-linearly with N .

interval required to give each node a collision-free TDMA slot. Extending to K simultaneous FL nodes requires $T_{\text{round}} \geq K \times T_{\text{slot}}$, which at $K=10$ would demand 100-second rounds, an impractically slow learning rate. Two paths exist: either switch to SF7 (recovering 14 dB link budget in exchange for roughly $4\times$ the channel capacity) or deploy the hierarchical topology we demonstrated, which reduces the number of nodes that must share a single TDMA channel.

The PA_BOOST misconfiguration is systematic. The Ra-02 module routes its antenna trace to the SX1276's PA_BOOST pin, not the RFO pin. The sandeepmistry/LoRa library defaults to RFO. This causes measured RSSI to be approximately -130 dBm at 2 m—indistinguishable from the -132 dBm noise floor—when the correct value should be -35 dBm. Every published LoRa project that uses the Ra-02 or similar Ai-Thinker module without explicitly setting PA_BOOST may have silently operated with 28 dBm less effective transmitted power than intended.

On-device SHA-256 is negligible. The 0.8 ms validation cost is 0.016% of the 5-second block cycle. Crucially, mbedTLS is already bundled with the ESP-IDF Arduino

TABLE VI
COMPARISON WITH RELATED SYSTEMS

System	Infra-free	On-MCU chain	On-device FL	Sensing
Dorri [17]	×	×	×	×
Zorbas [15]	×	×	×	IMU
Rahman [27]	×	×	✓	IMU
Nguyen [19]	×	×	✓	×
This work	✓	✓	✓	IMU+PPG

layer; blockchain integrity costs no additional code size compared to a system without it. The include path subtlety ("`mbedtls/md.h`" from the framework tree rather than as a `lib_dep`) is a one-line fix but burns several debugging hours in practice.

Comparison with gateway-dependent approaches. Table VI compares this work with representative prior systems.

Limitations. The two-node hardware testbed is small. While the Python simulation validates the system to $N=100$, the simulation's channel model uses log-normal shadowing rather than measured multipath profiles. Planned work includes multi-floor propagation measurements, energy profiling with an inline current logger, and extending the hardware testbed to four nodes to validate the hierarchical FL topology on physical radios.

VII. CONCLUSION

This paper presented a three-layer wearable monitoring system for first-responder teams that requires no external network infrastructure. Two physical ESP32-C3 nodes integrated LoRa P2P mesh, SHA-256 PoA blockchain, and FedAvg with hardware-entropy differential privacy on commodity hardware costing USD 12 per node. Key measured results are: PDR $\geq 98\%$ within 50 m at SF10; SHA-256 consensus latency of 0.38–0.85 ms from $N=10$ to $N=100$; 100% Byzantine rejection across 90 attack instances; TinyML fall detection AUC = 1.000; $10\times$ communication reduction from hierarchical FedAvg; and autonomous spoof-attack recovery in under 14 s.

The principal limitation is that TDMA FL scheduling at SF10 limits simultaneous FL-capable nodes on a shared channel to two before the round interval becomes impractical. Hierarchical aggregation resolves this by making only cluster heads contend for the shared LoRa channel, and the simulation validates the approach to 100 nodes. Future hardware work will extend the physical testbed to validate hierarchical FL on live radios and characterize energy consumption under sustained operation.

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